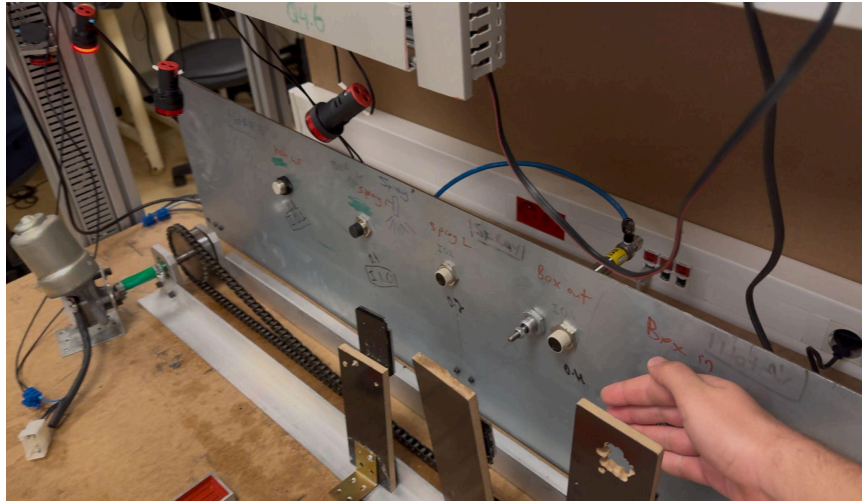


ME308 - Industrial Control

Lab #5 - Box Paint System

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Section - B



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Introduction

In this lab, we designed and programmed a PLC-based automation system to manage a box painting and drying process. The process involves:

1. Detecting box in and out locations while informing worker
2. Spray painting the box for 20 seconds.
3. Drying the paint in a hot air chamber for 1 minute.
4. Repeating this cycle three times before moving the box to the final storage location.

This system relies on sensors, actuators, and a PID temperature controller to ensure precise operation.

System Description

The system setup, as illustrated in Figure 1, includes the following components:

1. Actuators:

- **Two-Directional Motor (Q4.0, Q4.2):** Moves the conveyor belt to transfer the box between the spray chamber, hot air chamber, and the final storage area.
- **Spray Painting Gun (Q4.6):** Performs the painting operation for 20 seconds.
- **Hot Air Fan (Q4.6):** Blows hot air in the drying chamber for 1 minute.
- **Pneumatic Piston (Q4.4):** Pushes the finished box off the conveyor into storage.
- **Indicator Lamps:**
 - **Operation in Progress (Q4.7):** Lights up when the process is running.
 - **Ready for Operation (Q4.1):** Lights up when the system is ready for the next box.

2. Sensors:

- **Box In Sensor (I0.7):** Detects when a box is placed in the system at the starting position.

- **Box Out Sensor (I0.4):** Detects when the box reaches the end of the process.
- **Spray Left Sensor (I0.2):** Confirms when the box is inside the spray chamber.
- **Spray Right Door Sensor (I1.0):** Confirms when the box exits the spray chamber.
- **Hot Air Sensor (I0.1):** Detects when the box is inside the hot air chamber.
- **Start Button (I0.6):** Signals the system to begin the process.

Control Logic

1. The system starts when the operator presses the Start Button (I0.6) after placing a box at the Box In position (detected by Box In Sensor (I0.7)).
2. The conveyor, controlled by Motor Enable (Q4.0) and Motor Direction (Q4.2), moves the box to the Spray Chamber. The Spray Left Sensor (I0.2) confirms the box is properly positioned.
3. The Spray Painting Gun (Q4.6) performs the spray painting operation for 20 seconds.
4. The conveyor then moves the box to the Hot Air Chamber, confirmed by the Hot Air Sensor (I0.1). The Hot Air Fan (Q4.6) operates for 1 minute.
5. This cycle repeats three times before the conveyor moves the box to the Box Out position, detected by the Box Out Sensor (I0.4).
6. At the end of the process, the Pneumatic Piston (Q4.4) pushes the box off the conveyor into storage.

PID Controllers

Two PID controllers manage the temperature in the chambers:

I used 3 multiplication block and 2 addition block, both “Spray” and “Hot Air” used same PID block to adapt parameters. First multiplication consist of error and its constant k_p , second is multiplication of integral of error and k_i , third one is for derivative of error and k_d . These 3 products were added in two steps, this is because we can add a maximum of two values. (can be seen in figure 16-17)

- **Spray Chamber PID:** Parameters stored in memory words 30, 32, 34, and 36. Gains are:

- $K_p = 10$, $K_i = 1$, $K_d = 2$.
- **Hot Air Chamber PID:** Parameters stored in memory words 40, 42, 44, and 46.
Gains are:
 - $K_p = 20$, $K_i = 1$, $K_d = 3$.

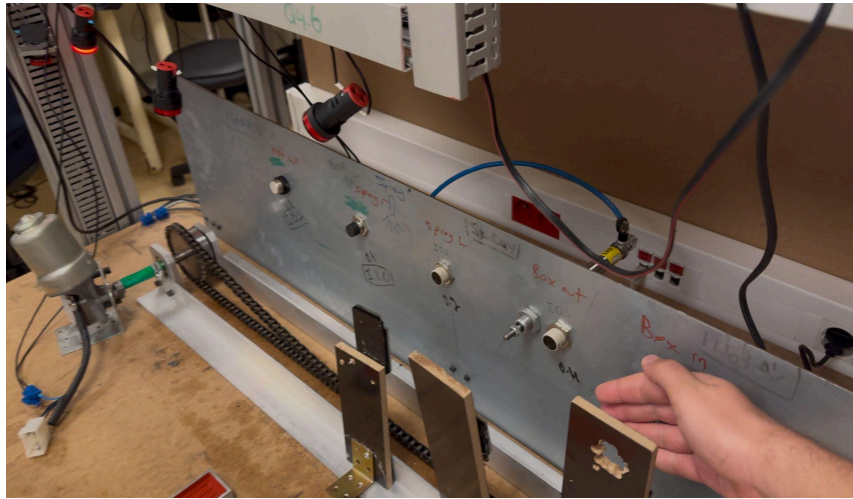


Figure 1: General Setup

PLC Program Details

The ladder logic program was designed using Siemens TIA Portal to manage the spray painting and hot air drying process. The program consists of the following networks:

Network 1: System Ready State

- **Purpose:** The system waits for the operator to place the box at the Box In position and press the Start Button (I0.6). (can be seen in figure 2)
- **Inputs:**
 - **Box In Sensor (I0.7):** Detects the box.
 - **Start Button (I0.6):** Signals the system to start.
- **Outputs:**
 - **Ready for Operation Lamp (Q4.1):** Lights up when the system is ready.
 - **Operation in Progress Lamp (Q4.7):** Turns on when the process begins.

Network 2: Spray Painting Process

- **Purpose:** Activates the spray painting operation when the box reaches the spray chamber. (can be seen in figure 3)
- **Inputs:**
 - **Spray Left Sensor (I0.2):** Confirms the box is in position.
- **Outputs:**
 - **Spray Painting Gun (Q4.6):** Simulates the spray painting operation for 20 seconds.
- **Timer:**
 - A 20-second timer is used to control the duration of the spray. (in the figure 3, timer is set to 2 seconds in order to test and observe the system faster.)

Network 3: Conveyor Movement to Hot Air Chamber

- **Purpose:** Moves the box to the hot air chamber after spray painting. (can be seen in figure 4)
- **Inputs:**
 - **Spray Right Door Sensor (I1.0):** Confirms the box has exited the spray chamber.
- **Outputs:**
 - **Motor Enable (Q4.0) and Motor Direction (Q4.2):** Move the conveyor forward.

Network 4: Hot Air Drying Process

- **Purpose:** Activates the hot air drying process when the box reaches the hot air chamber. (can be seen in figure 5)
- **Inputs:**
 - **Hot Air Sensor (I0.1):** Confirms the box is inside the hot air chamber.
- **Outputs:**
 - **Hot Air Fan (Q4.6):** Simulates the drying operation for 1 minute.
- **Timer:**
 - A 1-minute timer controls the duration of the hot air operation. (in the figure 5, timer is set to 6 seconds in order to test and observe the system faster.)

Network 5: Manual Cycle Handling

- **Purpose:** Ensures the spray and hot air operations repeat three times. This is done by manually programming three sets of spray and hot air cycles.
- **Logic:**
 - Each spray and hot air operation is repeated by duplicating the relevant ladder logic for each cycle.

Network 6: Box Out and Repeat

- **Purpose:** Moves the box to the storage area after completing all cycles. The system becomes ready again for another box. (can be seen in figure 6-7)
- **Inputs:**
 - **Box Out Sensor (I0.4):** Detects the box at the final position.
- **Outputs:**
 - **Pneumatic Piston (Q4.4):** Pushes the box into storage.
 - **Operation in Progress Lamp (Q4.7):** Turns off when the process is complete.

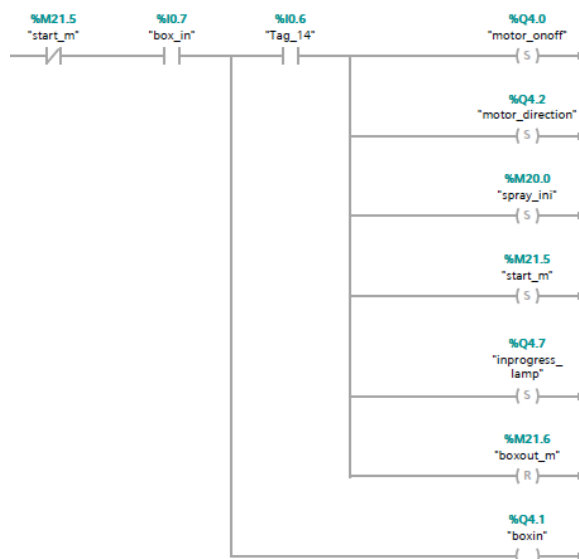


Figure 2: Network 1 PLC Program

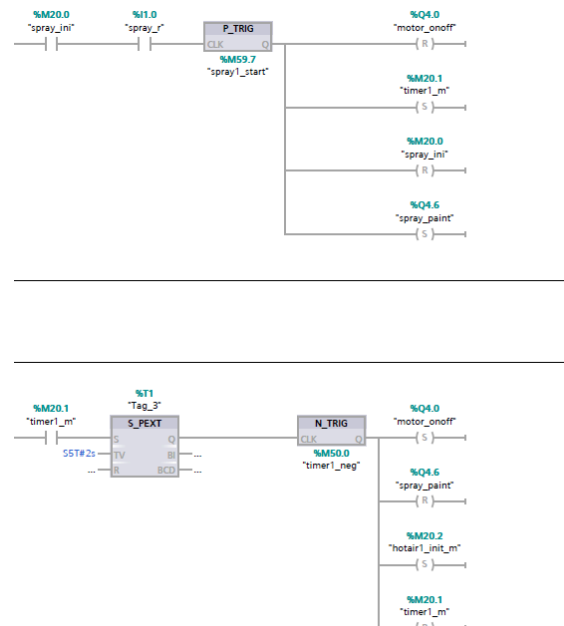


Figure 3: Spraying Part of PLC Program

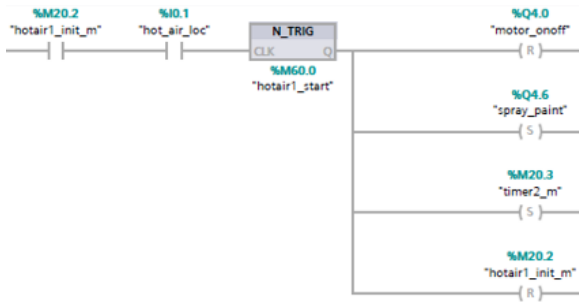


Figure 4: Hot Air Part of PLC Program

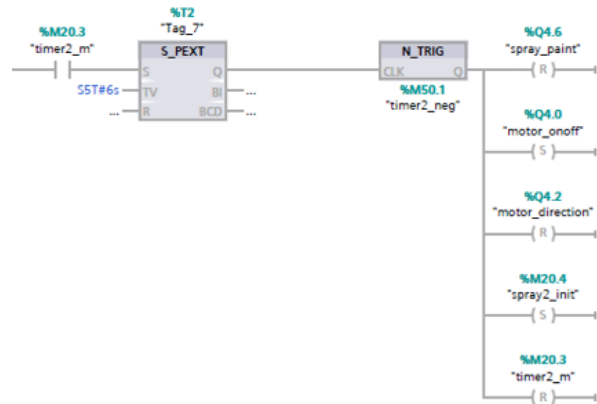


Figure 5: Hot Air Timer of PLC Program

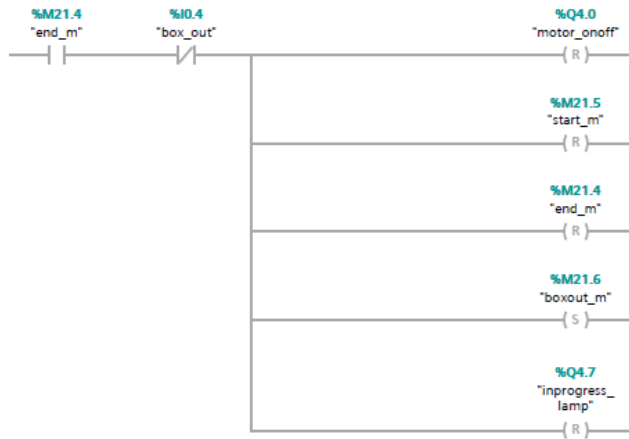


Figure 6: End Program and Repeat Program Part of PLC Program



Figure 7: Box Out Part of PLC Program

Execution and Results

1. System Initialization:

- When the operator places the box at the Box In position and presses the Start Button, the system indicates readiness with the Ready for Operation Lamp. (can be seen in figure 8-9)
- The process begins, and the Operation in Progress Lamp lights up.

2. Spray Painting:

- The conveyor moves the box to the spray chamber. Once detected by the Spray Left Sensor, the Spray Painting Gun operates for 20 seconds.
- The process is visually represented by the Spray/Hot Air Lamp (Q4.6). (can be seen in figure 10-11)

3. Hot Air Drying:

- After spray painting, the conveyor moves the box to the hot air chamber. When the box is in position, the Hot Air Fan operates for 1 minute. (can be seen in figure 12)

4. Cycle Repetition:

- The spray and hot air operations are repeated manually three times by duplicating the logic. (can be seen in figure 13)

5. Box Out and Storage:

- After completing three cycles, the conveyor moves the box to the Box Out position. The Pneumatic Piston pushes the box into storage. (can be seen in figure 14)
- The Operation in Progress Lamp turns off, and the system resets for the next box. (can be seen in figure 15)

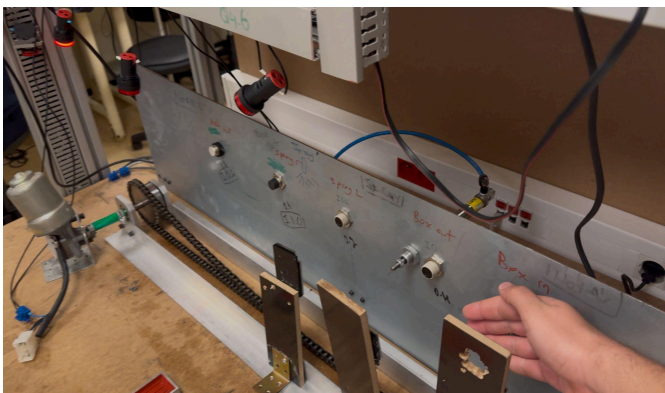


Figure 8: Box is in position.



Figure 9: Start Button Pressed

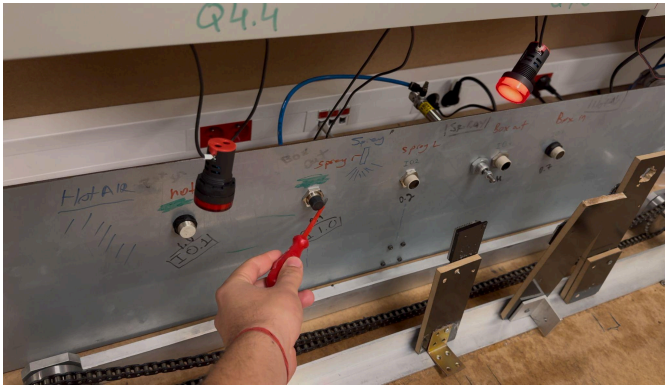


Figure 10: Box comes spray position and spray starts.



Figure 11: Box starts moving after 20 seconds.

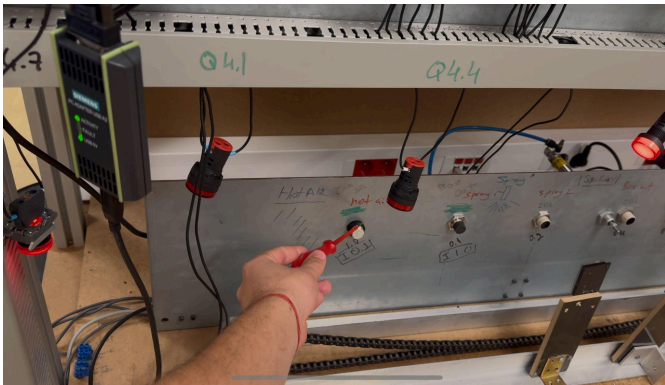


Figure 12: Box in hot air position, Hot air starts

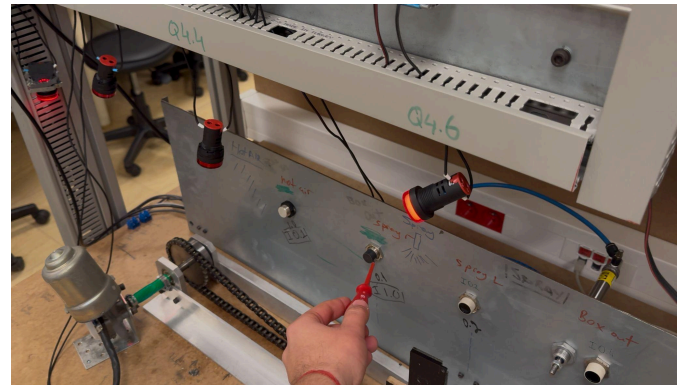


Figure 13: Box goes back to spray position and 3-repeat cycle starts.



Figure 14: Box is in out position. Out signal on.



Figure 15: General cycle is ready again to start.

Discussion

This lab showed how a PLC can control a multi-step process, like spray painting and drying, in an automated way. Overall, the system worked as expected, and also demanded results were obtained from the PID controller.

System Performance

1. Sequential Operation:

The system correctly followed the steps for painting, drying, and moving the box. Sensors like the Box In Sensor (I0.7) and Hot Air Sensor (I0.1) worked accurately to detect the box at the right positions. This made sure the spray and drying processes happened at the right time and place.

2. Shared Outputs (Spray and Hot Air Lamp):

We used the same output (Q4.6) to represent both the spray painting and hot air operations. While this saved hardware, it made it harder to tell which process was running. Having separate lamps for spray and hot air would make it clearer for the operator.

3. Manual Cycle Handling:

Instead of using a counter, we manually repeated the spray and hot air logic for three cycles. This worked fine for this lab, but it made the ladder logic longer and harder to manage. If more cycles were needed, using a counter would make the program simpler and easier to adjust.

4. Box Out and Storage:

The Pneumatic Piston (Q4.4) successfully pushed the box into storage after the process was done. The Box Out Sensor (I0.4) worked well to detect when the box reached the final position. This ensured the system stopped at the right time and got ready for the next box.

PID Controllers

The PID controllers (can be seen in figure 17) were important for keeping the temperatures of the spray and hot air chambers under control. Here's how they worked and what could be improved:

1. PID Setup:

We used two separate PID controllers:

- **Spray Chamber PID:** (can be seen in figure 16-18)
 - Gains: $K_p = 10$, $K_i = 1$, $K_d = 2$
 - Stored in memory words 30, 32, 34, and 36.
 - This controller ensured a steady temperature for the spray chamber, which doesn't require rapid changes.
- **Hot Air Chamber PID:** (can be seen in figure 16-19)
 - Gains: $K_p = 20$, $K_i = 1$, $K_d = 3$
 - Stored in memory words 40, 42, 44, and 46.
 - The higher proportional gain (K_p) was needed because the hot air chamber experiences bigger temperature changes.

2. Temperature Control:

The PID controllers kept the temperatures steady during the process.

- The **Ki (integral gain)** helped fix any small temperature errors over time.
- The **Kd (derivative gain)** smoothed out sudden changes, so the system didn't overshoot or undershoot the target temperature.

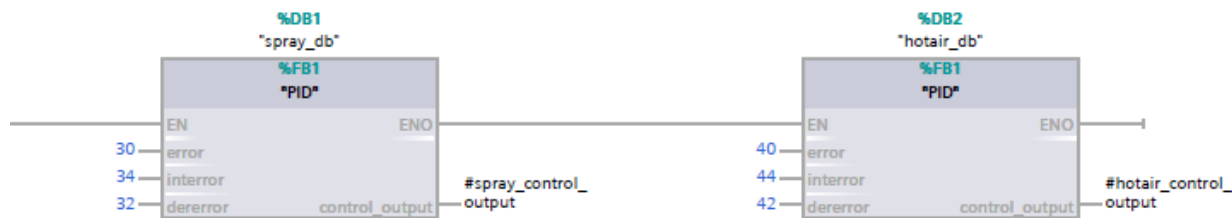


Figure 16: Both data blocks' implementation on PID FB in main function..

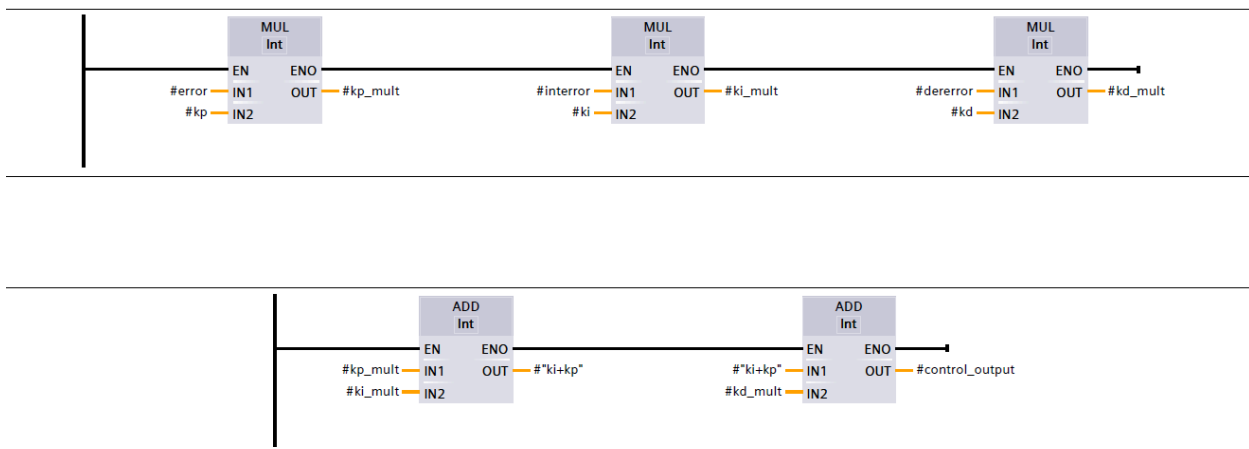


Figure 17: PID Function Block

spray_db			
Name	Data type	Offset	Start value
▼ Input			
error	Int	0.0	0
interror	Int	2.0	0
dererror	Int	4.0	0
▼ Output			
control_output	Int	6.0	0
InOut			
▼ Static			
kp	Int	8.0	10
kd	Int	10.0	2
ki	Int	12.0	1

Figure 18: Spray Chamber Data Block

hotair_db			
Name	Data type	Offset	Start value
▼ Input			
error	Int	0.0	0
interror	Int	2.0	0
dererror	Int	4.0	0
▼ Output			
control_output	Int	6.0	0
InOut			
▼ Static			
kp	Int	8.0	20
kd	Int	10.0	3
ki	Int	12.0	1

Figure 19: Hot Air Chamber Data Block